

Supporting information for: "Developing a modular and parametrized framework for constructing metal life cycle inventories: application to Canada"

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Selection of facilities from the MetalliCan dataset

Figure 1 presents the key decision points that determine whether a site qualifies for a site-specific LCI.

Figure 2 presents an overview of the 35 selected sites, providing the province, mining

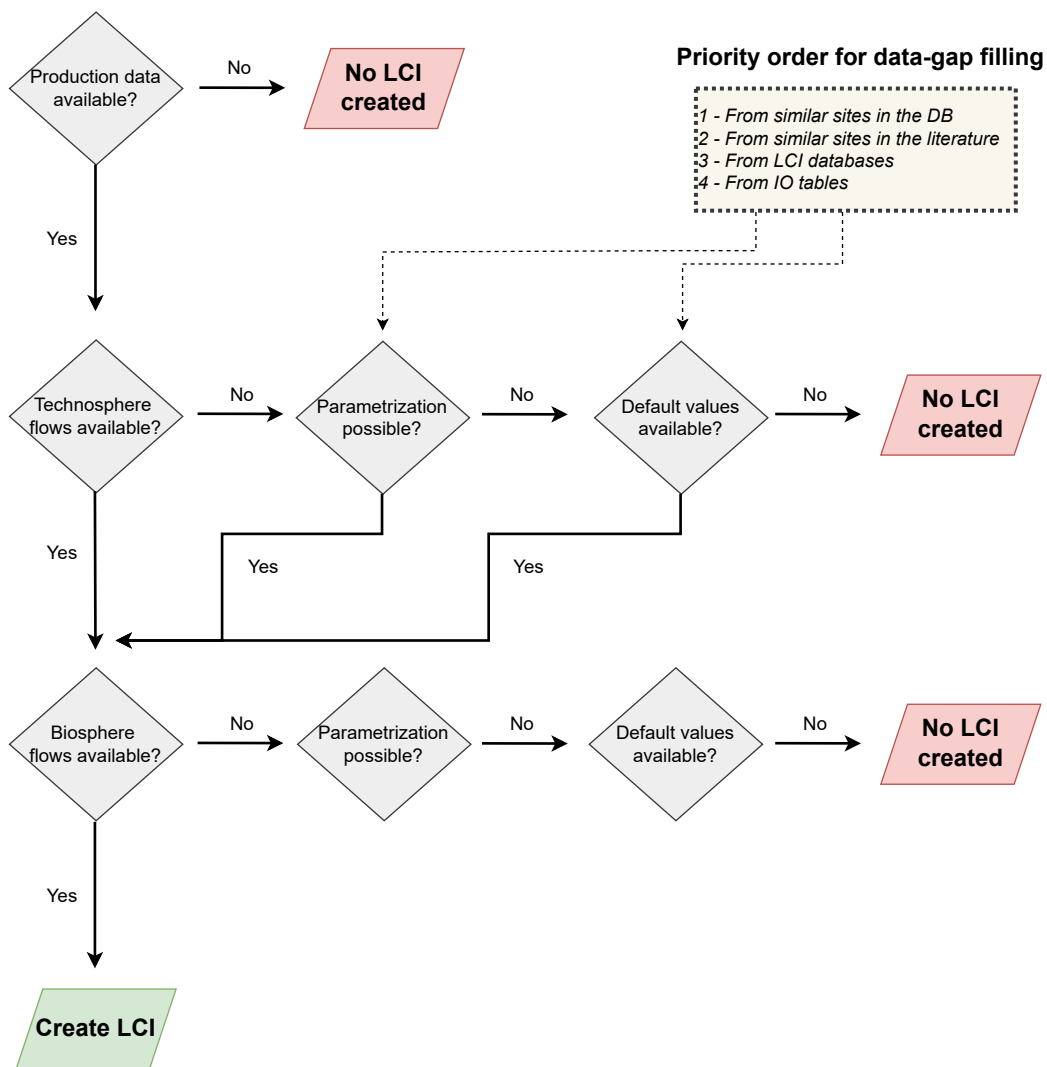


Figure 1: Decision tree for site-specific LCI creation from MetalliCan dataset and choices of additional sources

method, archetype and production types.

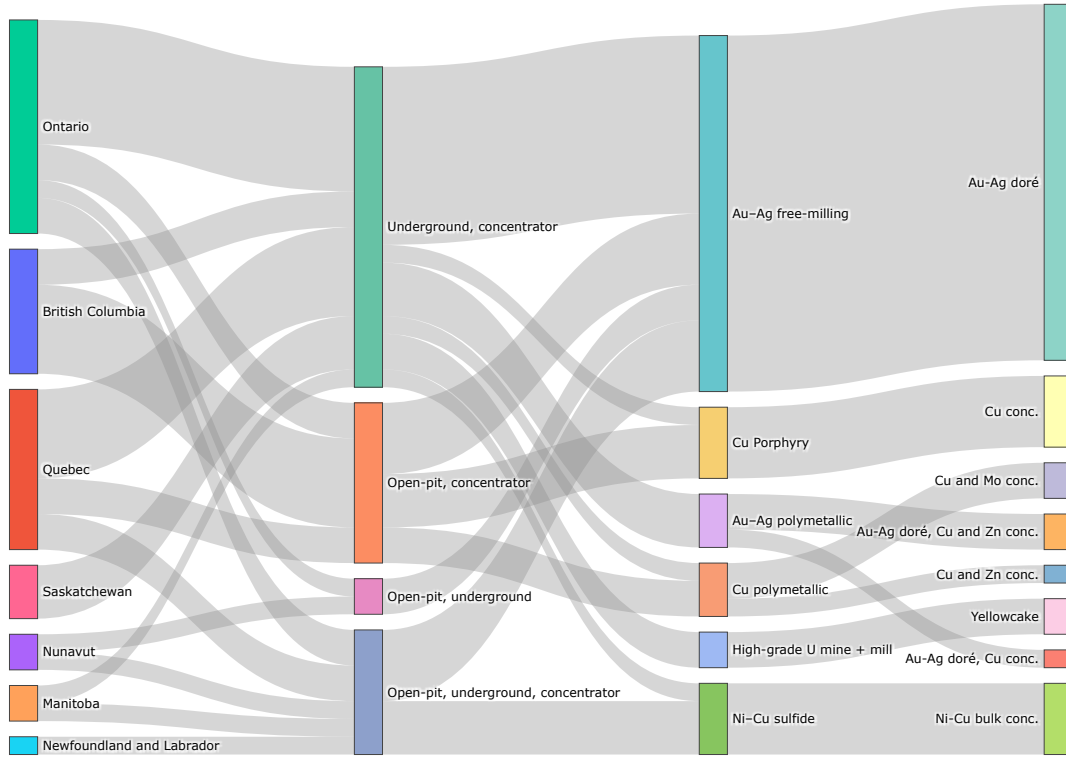


Figure 2: Province, mining method, archetype and production of selected sites

Detailed information on the specific sites, including geographic location and production volumes, are given in a spreadsheet format in 'SI-2-site-selection.xlsx'.

Sources for data-gap filling

Detailed data are given in a spreadsheet format in 'SI-3-data-gap-filling.xlsx'.

Parametrization and default values from the literature

In LCA, parametrization is used to express key inventory flows as functions of underlying operational variables such as in the mining context ore grade, mining depth, energy mix, or recovery rates rather than as fixed point values. This approach increases transparency, because model assumptions are explicitly documented, and enhances flexibility, as inventories

can be recalculated dynamically for different sites or scenarios.¹ When parameter ranges are uncertain or variable, Monte Carlo sampling can be applied to explore the resulting distribution of outcomes and assess which parameters most influence environmental performance.

Parametrization in the mining and metallurgical LCA literature has been predominantly driven by energy–ore grade relationships and mining methods, e.g. open-pit vs underground. Most studies focus on how declining ore grades and increasing extraction challenges modify energy intensity, greenhouse gas emissions, and water requirements, often modelling energy use as a function of ore grade and operational configuration. Several studies, focusing on one metal, e.g. gold,^{2,3} copper,^{4,5} nickel,⁶ uranium,⁷ or several metals,^{8,9} show that energy use increase exponentially as ore grade declines, particularly below a technical threshold where comminution and processing efficiencies decrease sharply. This relationship is also sometimes demonstrated for GHG and water.^{2,5} Another consistent finding is that open-pit operations rely more heavily on diesel due to the handling of large volumes of waste and ore, whereas underground mines depend more on electricity, particularly for ventilation, dewatering, hoisting, and ground support installation. Mining depth is also cited as an important driver, electricity demand in underground mines increases log-linearly, with estimates of roughly 7% additional electricity per 100 m depth.⁴ Some studies also typically split energy consumption between steps, e.g. mining and beneficiation, assigning an energy mix to each step, e.g. Argonne for copper and nickel.¹⁰ Some studies attempt to integrate parametric works along all steps, from mining to refining. For example, Manjong et al.¹¹ evaluate multiple levers—ore grade, mining method, mineral composition, and reagent concentrations—to explain variability in the environmental impacts of battery raw materials, while Gibon et al.⁷ do similar things for nuclear plants. Roy et al.¹² propose a mass-based scaling factor that adjusts mining and beneficiation inventories based on the total mass of rock mined, using strip ratio and ore grade to rescale flows while keeping the mining method and ore type constant.

Overall, parametrization remains overwhelmingly centered on energy consumption, ore

grade, and mine depth. Other technosphere flows—such as reagent use, grinding media, explosives, ground support materials, backfill binders, and water treatment chemicals—remain far less explored, despite their importance for metal-specific environmental profiles. This gap underscores the need for more comprehensive, multi-parameter approaches capable of capturing the full process variability across mining and metallurgical operations.

Parametrization and default values from MetalliCan

The following graph represents land occupation as a function of ore processed. All the code to generate the graphs is available on Github at: https://github.com/marpellan/metallican_lci

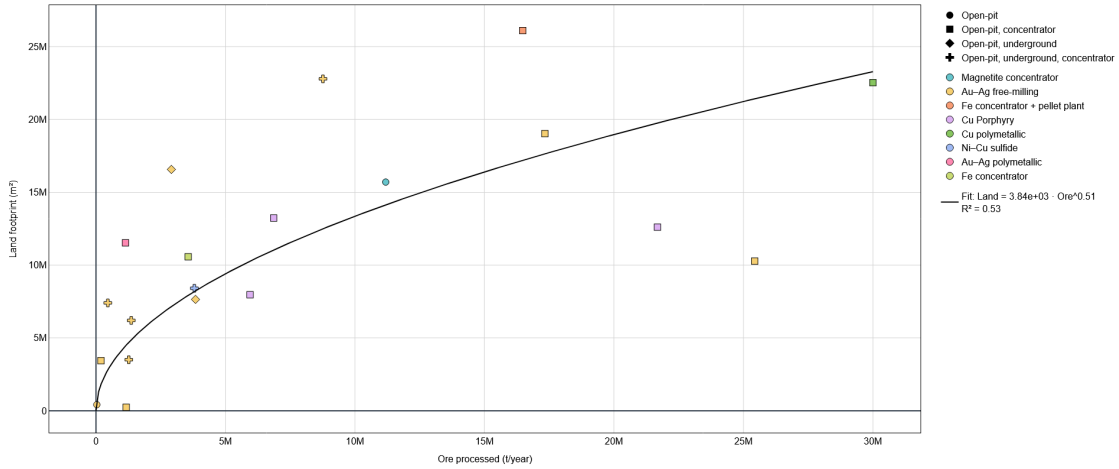
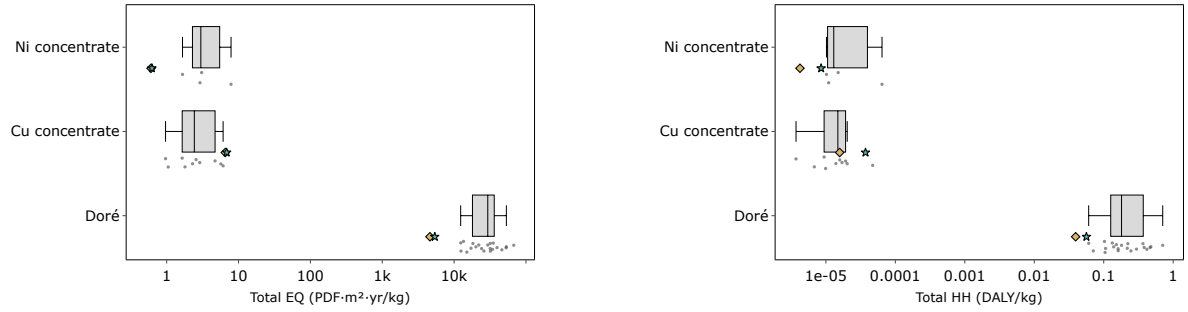


Figure 3: Land use ore processed relationship

Additional results

The following figures are complementary to Figure 3,4 and 5 in the main manuscript and provides similar results applied to particulate matter and land transformation impacts.

Site-variability for total endpoints damages



((a)) Total ecosystem quality (EQ) damages across reference products

((b)) Total human health (HH) damages across reference products

Figure 4: Total site-level damages for ecosystem quality (EQ) and human health (HH) across reference products. Boxplots show the median, interquartile range, and whiskers extending to $1.5 \times$ the interquartile range.

□ MetalliCan range • MetalliCan sites ♦ Ecoinvent ★ Regioinvent

Direct and indirect particulate matter and land transformation contribution from primary and inferred inventory data

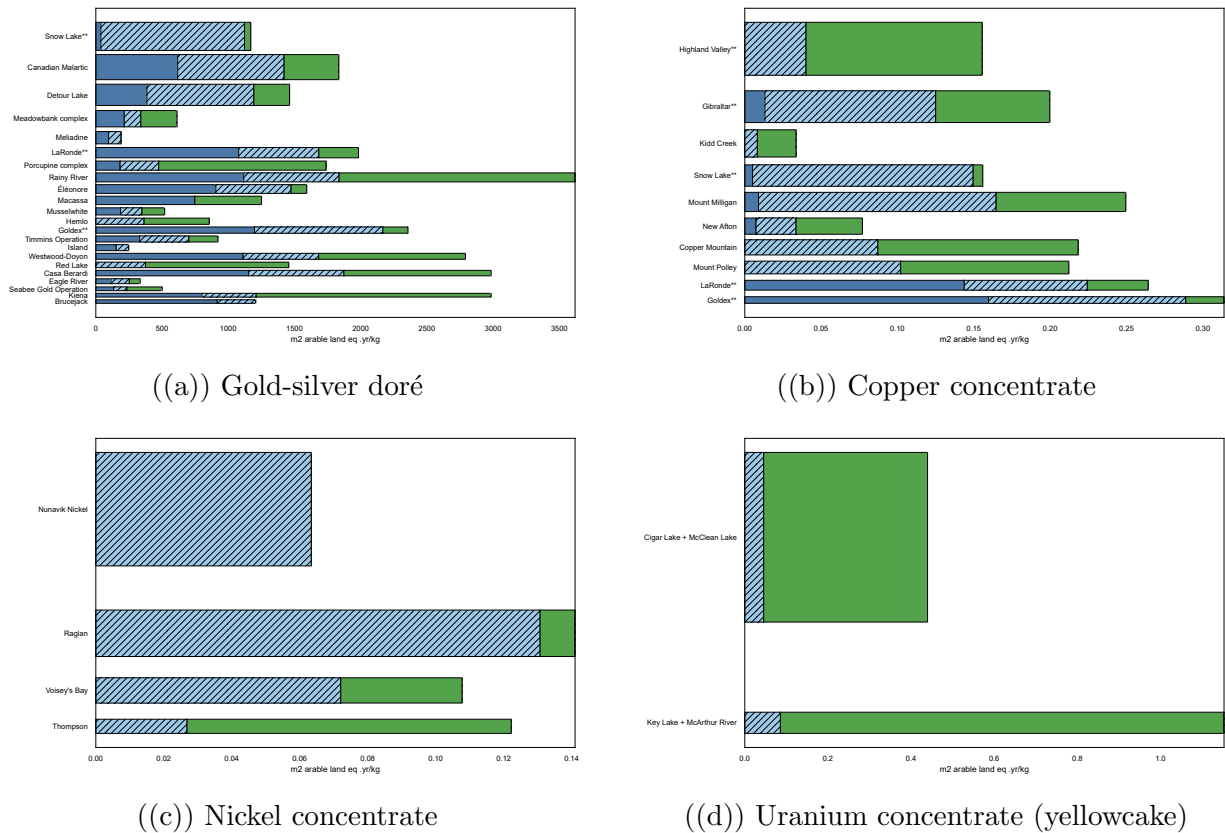
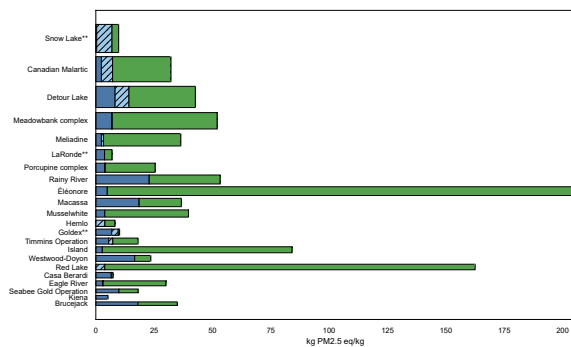
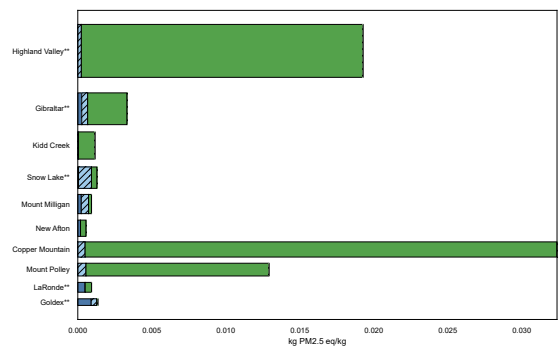


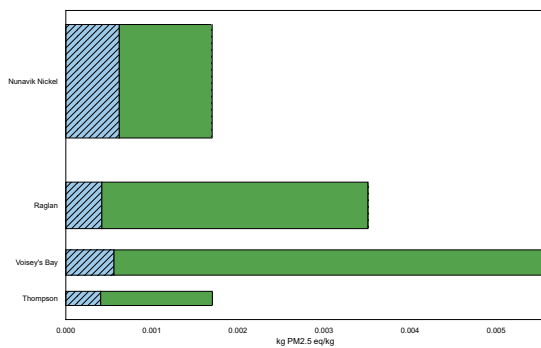
Figure 5: Direct, indirect, primary and inferred contribution in total land transformation impacts, ranked by production volume in descending order



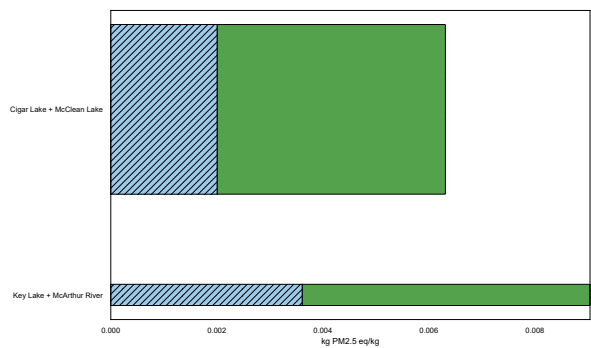
((a)) Gold-silver doré



((b)) Copper concentrate



((c)) Nickel concentrate



((d)) Uranium concentrate (yellowcake)

Figure 6: Direct, indirect, primary and inferred contribution in total particulate matter impacts, ranked by production volume in descending order

Particulate matter and land transformation impacts across sites and commodities

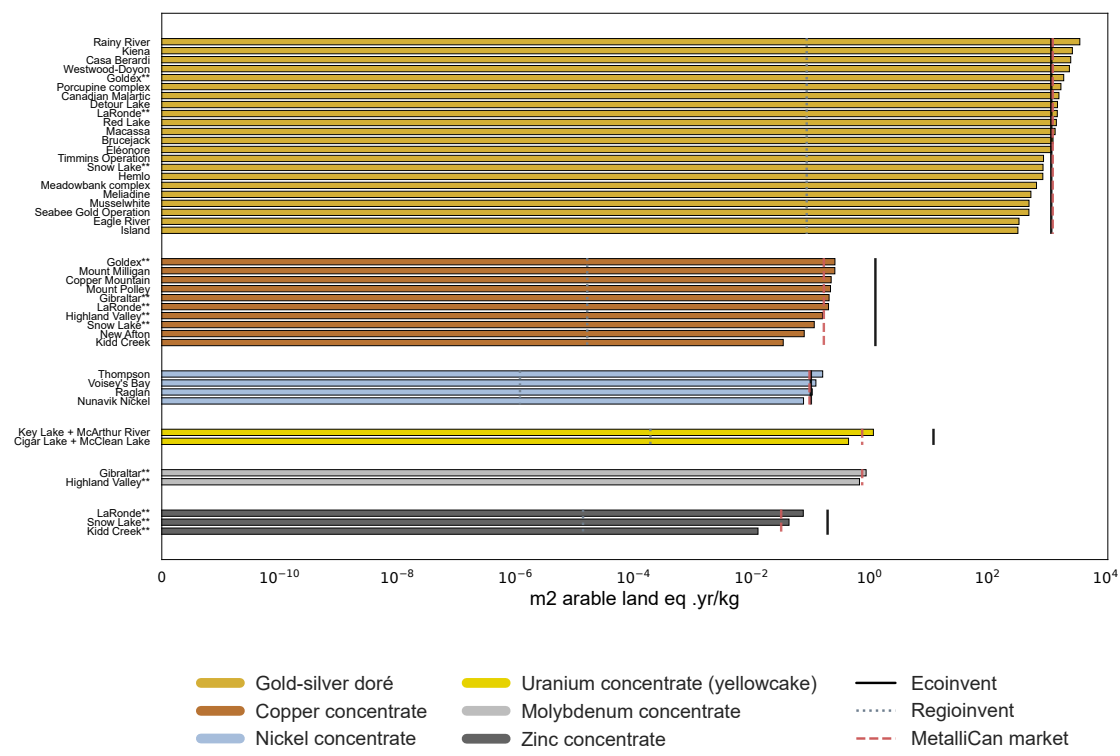


Figure 7: Difference between sites and commodities in total land transformation impacts with ecoinvent and Regioinvent proxies.

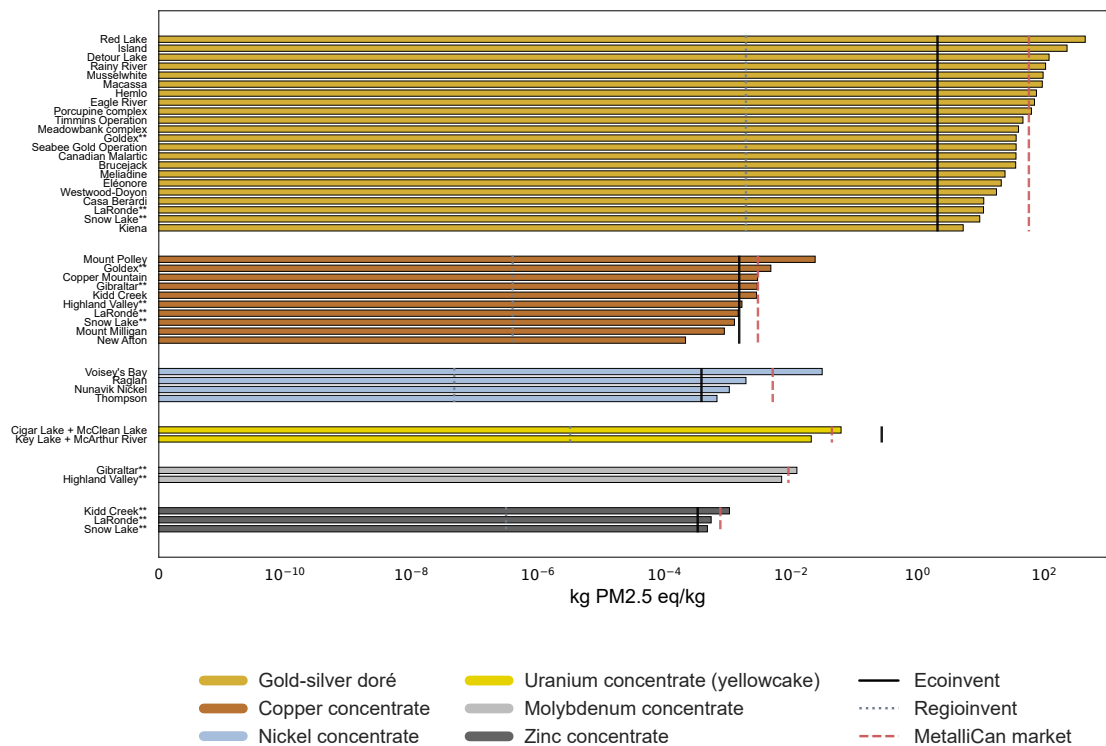


Figure 8: Difference between sites and commodities in total particulate matter impacts with ecoinvent and Regioinvent proxies.

Contribution analysis for particulate matter and land transformation

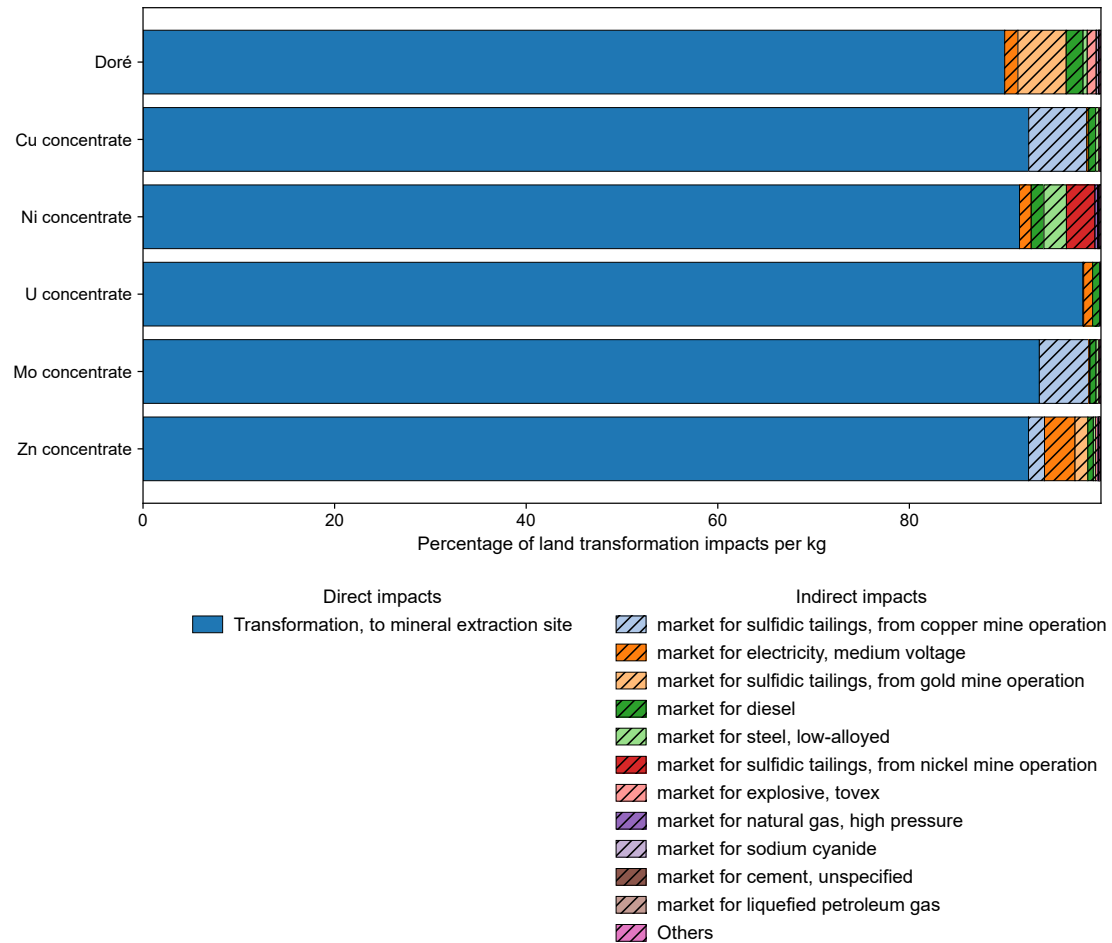


Figure 9: Land transformation contribution analysis for market-weighted MetalliCan LCIs per commodity

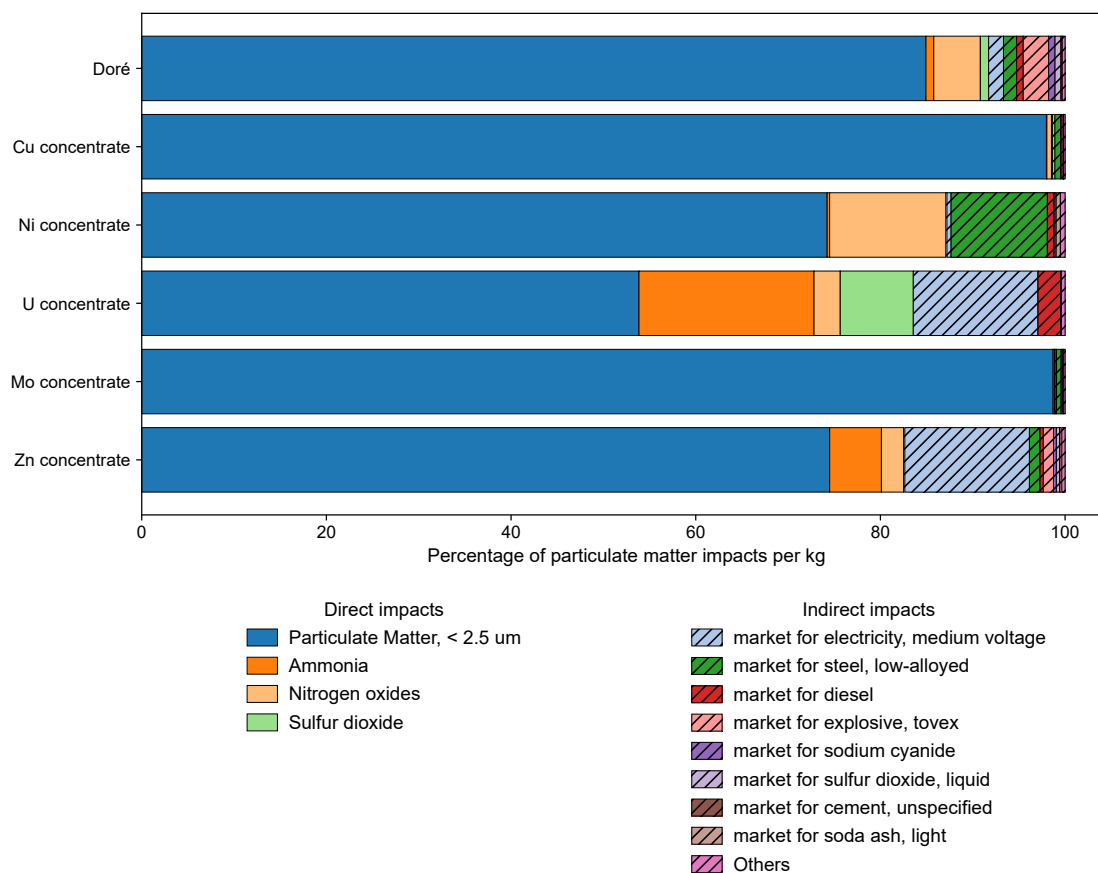


Figure 10: Particulate matter contribution analysis for market-weighted MetalliCan LCIs per commodity

Update of the climate change damage indicator (ecosystem quality) in Impact World+ version 2.2.

One of the most influential changes between IMPACT World+ versions 2.1 and 2.2 concerns the climate change damage indicator for ecosystem quality. In version 2.2, both fate and effect factors were revised. A key conceptual change is the explicit inclusion of marine species alongside terrestrial species in the affected taxa.¹³ As a result, climate change impacts on ecosystem quality are now reported through separate terrestrial and marine indicators, which can be aggregated to estimate total ecosystem damage. The combined effect factor in version 2.2 is approximately 13 times higher than in version 2.1. Consequently, climate change

becomes the dominant contributor to ecosystem quality damages for most human activities, substantially increasing its relative contribution compared to results obtained using version 2.1.

As a result, when using Impact World+ version 2.2., climate change represent more than 95% on total ecosystem quality damages, when compared to results presented in the main manuscript in Figure 2.

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